•Fiber Optic Network Design

College of Opt. Sci. & Engr., ZJU

2026

•Outline

- •Future Optical Amplifier
- •Design of Hybrid Amplifier
  - Gain Optimization
  - Gain Balance
  - Noise Figure
- •Exercise

What's the Next for Optical Amplifiers?

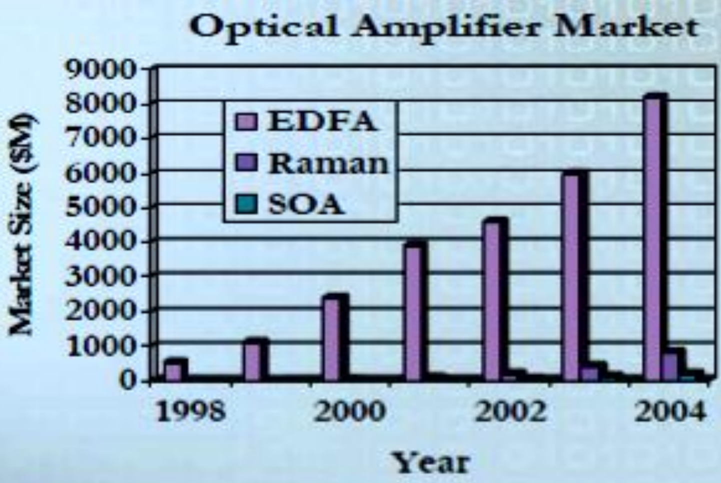
“Prediction is very difficult, especially if it’s about the future.”

Niels Bohr
1922 Nobel Prize in Physics

Intelligent EDFAs

Initial focus: optical amplifiers

- Periodically amplify optical signals which lose power due to optical fiber attenuation
- A key enabler for WDM technology: amplifies all channels within the gain band
- Arguably the most critical sub-system in optical communication networks



Pricing of EDFA modules

Year 2000 - $20K-40K

Year 2005 - $4K-6K

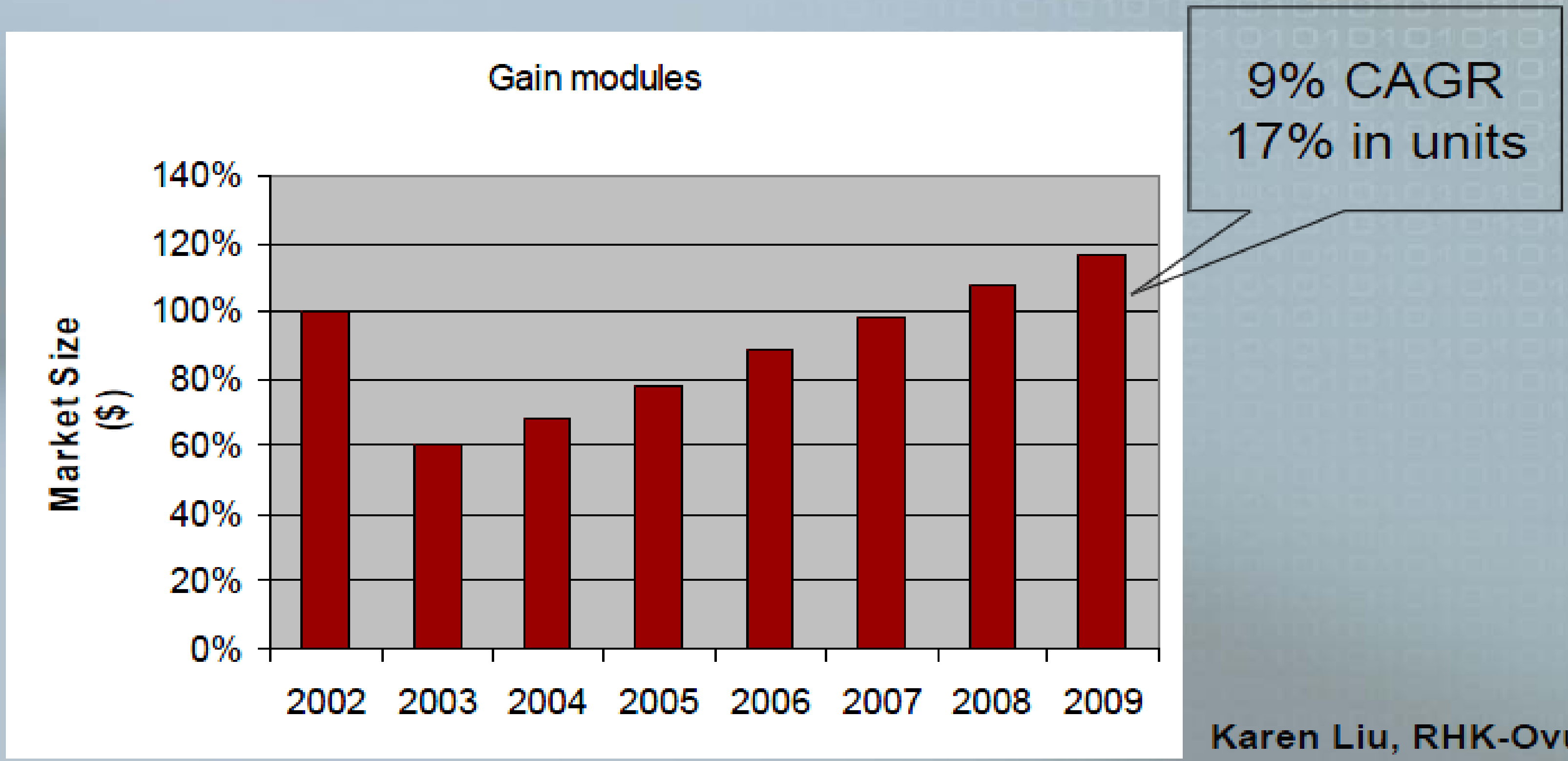
Novel EDFA: The First Product

- Exploding market:: $1B, 52% CAGR
- New system players with new requirements: Corvis, Qtera, ONI, Sycamore
- New features: Integration with Raman; bi-directional transmission; transient control; etc.
- New devices ready: Dynamic gain equalization filter (DGEF); Tunable dispersion compensator (TDC); High power pump, etc.

Raman Optical Amplifier (ROA): The Second Product

- New market: No commercial products on the market
- Benefits: Distributed Raman - OSNR improvement; Lumped Raman - New wavelength operation range
- The technology is ready: High power pumps are available.
- S-band ROA

EDFA is back to normal through crash



Karen Liu, RHK-Ovum

Long-haul optical building has resumed, metro build also strong

The "next-generation" is finally here with Ethernet, IP everywhere as well as other associated "new" technologies

Hybrid Amplifiers

Overview of Raman and EDFAs Combined Amplification

Why use Hybrid Amplifiers

- Enlarges the transmission capacity of broadband systems
- "Upgrading" the existing systems built with EDFA amplifiers with broader/flatter bandwidth
- Ability to carry more wavelength-multiplexed optical channels at given spacing among the channels
- Raman amplification gives flexibility to the selected band amplification

Basic Idea of Hybrid Amplifiers

Principle of working / Operating Mode

- Flatten EDFA gain by using Raman gain
- Improve gain performance at longer signal wavelengths

Typical Configurations

Raman Amplifiers setup in different configurations:

- Discrete
- Distributed
- backward-, forward- and bidirectional pumped
- numerous types of fibers (DSF, DCF to be used as SRS active media)

EDFAs setup considers:

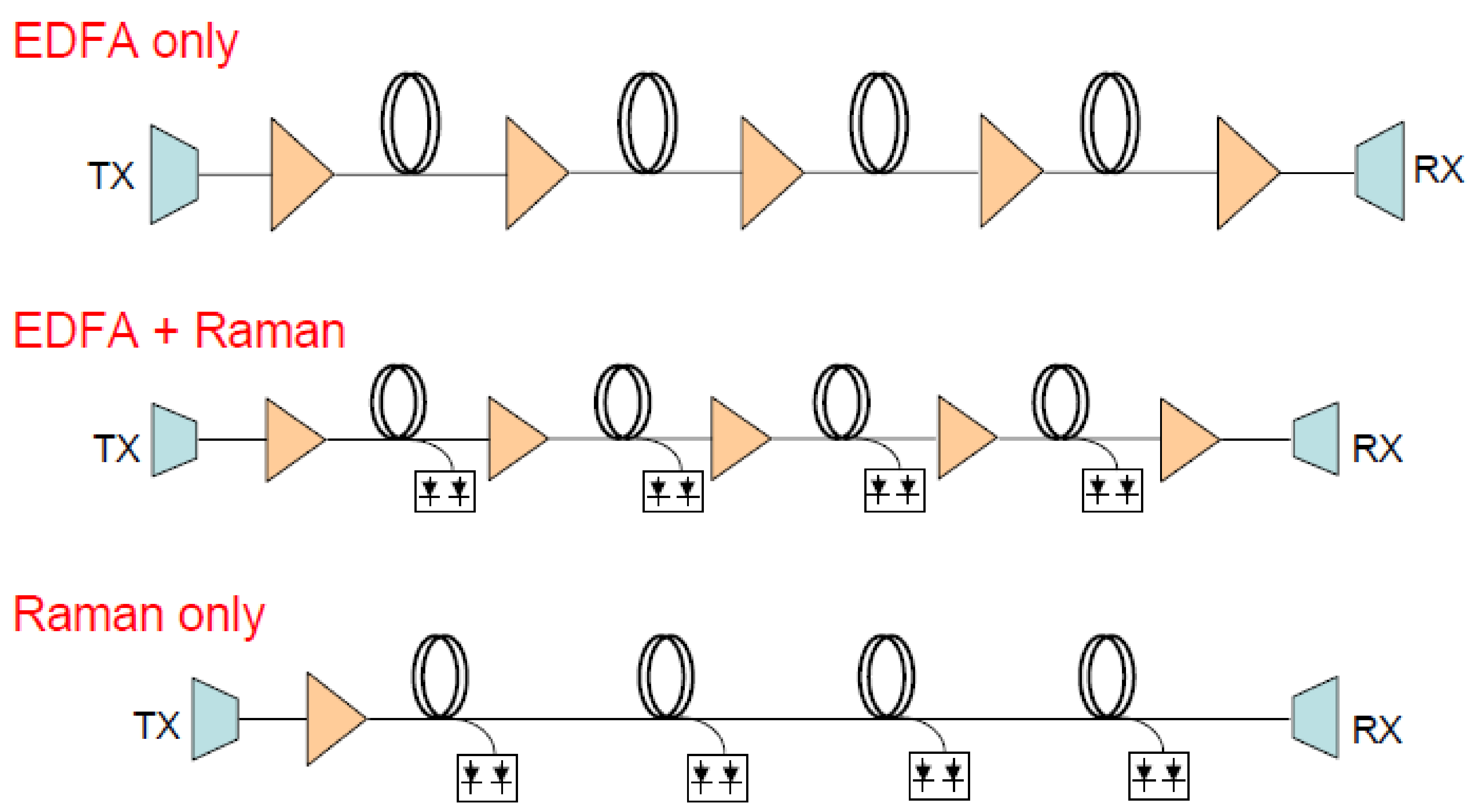
- series or parallel configuration
- single or multiple fiber stages
- C- and L-band regions

Hybrid Amplifiers Issues

The Hybrid Amplifiers studies are concerned with:

- **maximizing the span length and/or minimizing the impairments of fiber nonlinearities**
- **enhancing the EDFAs' bandwidth**
- **designing "optimal" hybrid amplifiers in order to obtain flat and widest output gain performance**

Comparing Different System Configuration



Results Obtained After Comparing the 3 Types of Systems

Considering in-line and pre-amplification functions:

- **<u>Long-haul EDFA-only systems：</u> limited by OSNR and NL effects**
- **<u>Raman-only systems：</u> limited by SNR caused by double Rayleigh backscattering**
- **<u>Combination of distributed Raman and EDFAs：</u> present better performance than conventional EDFA-only systems**

Gain Balance Between Raman and EDFAs

- Complex problem with several degrees of freedom

  \- optimization technique
- Main considerations to determine the optimum gain balance between Raman and EDFAs
- Most important parameters
  - OSNR, gain-flatness, bandwidth
  - number of channels, number of spans,
  - Maximum transmission capacity

Techniques that Enlarge Flattened Gain-Bandwidth of "Discrete" Fiber Amplifiers

I. New host materials

- fluoride- and telluride-based EDFAs
- thulium-doped fiber amplifier

II. Using EDFAs with GEQ + discrete Raman amplifier

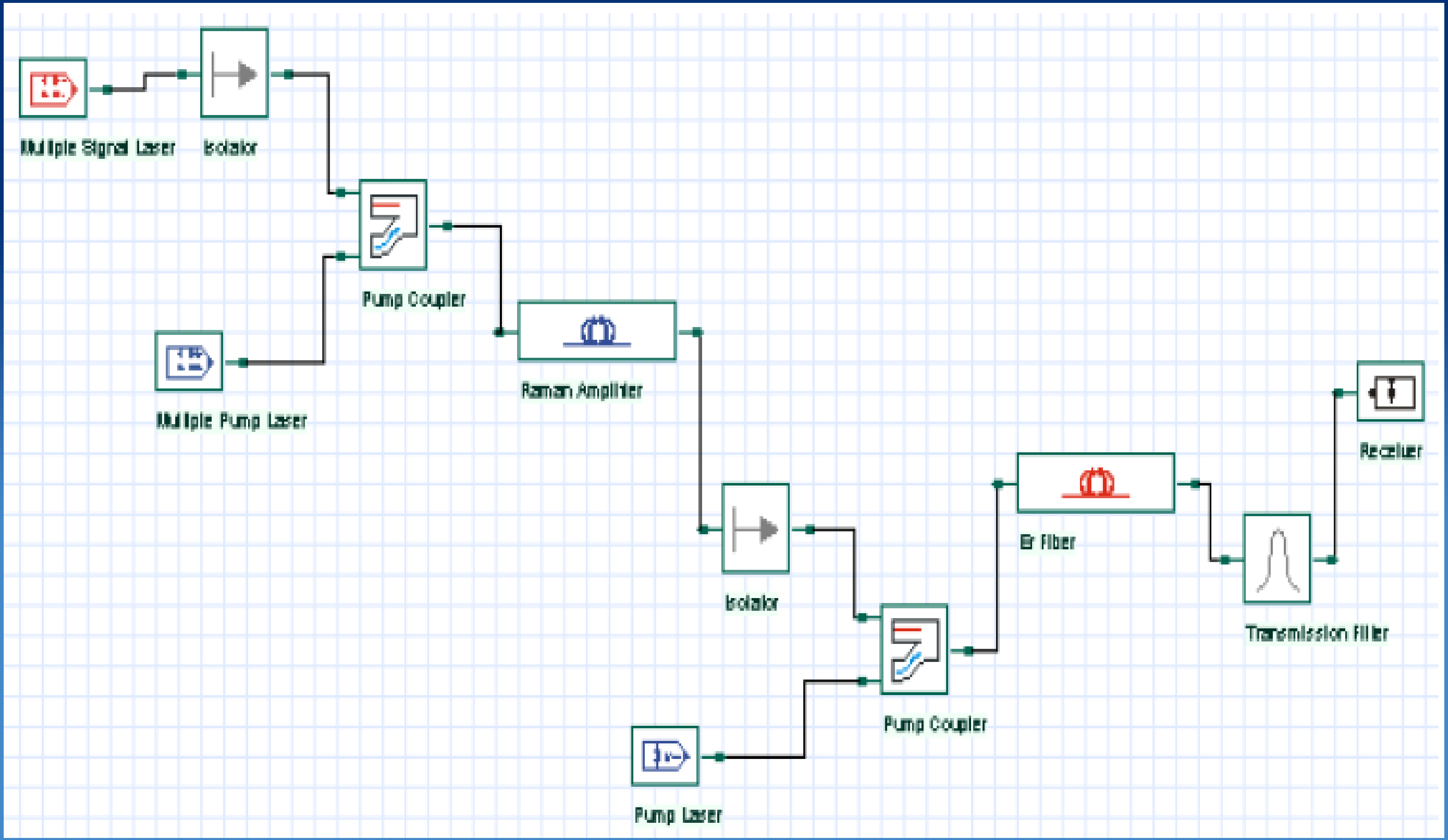
III. Different amplifier configurations

- two-gain band
- parallel/series configuration – multiple fiber stage
- gain-equalizing (GEQ) filters

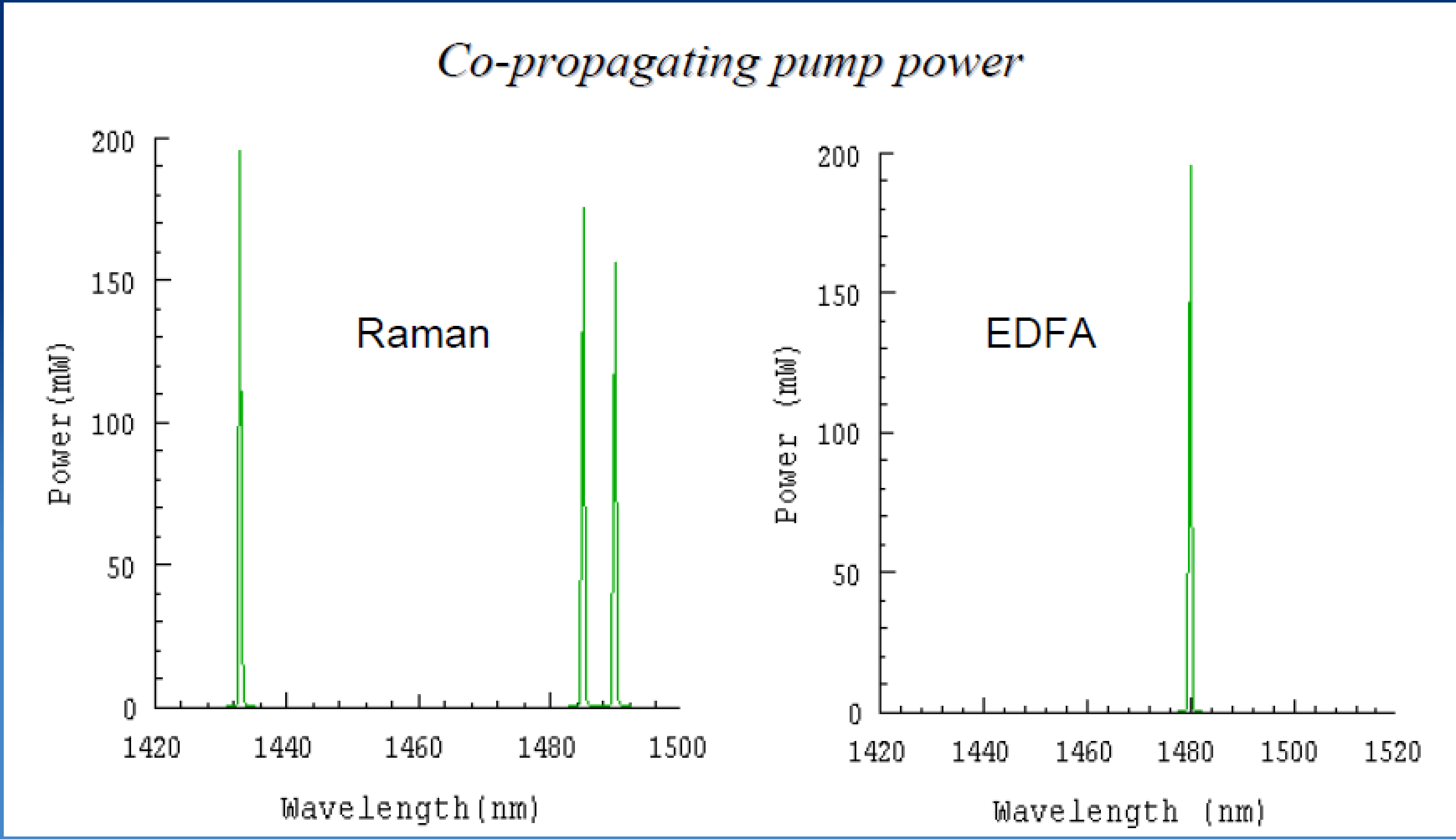
Typical Values Observed in Hybrid Amplifiers

- Bandwidth > 40 nm
- Gain 15 – 25 dB
- Gain Ripple < 11.3 %
- Noise Figure < 6 dB
- OSNR > 32 dB

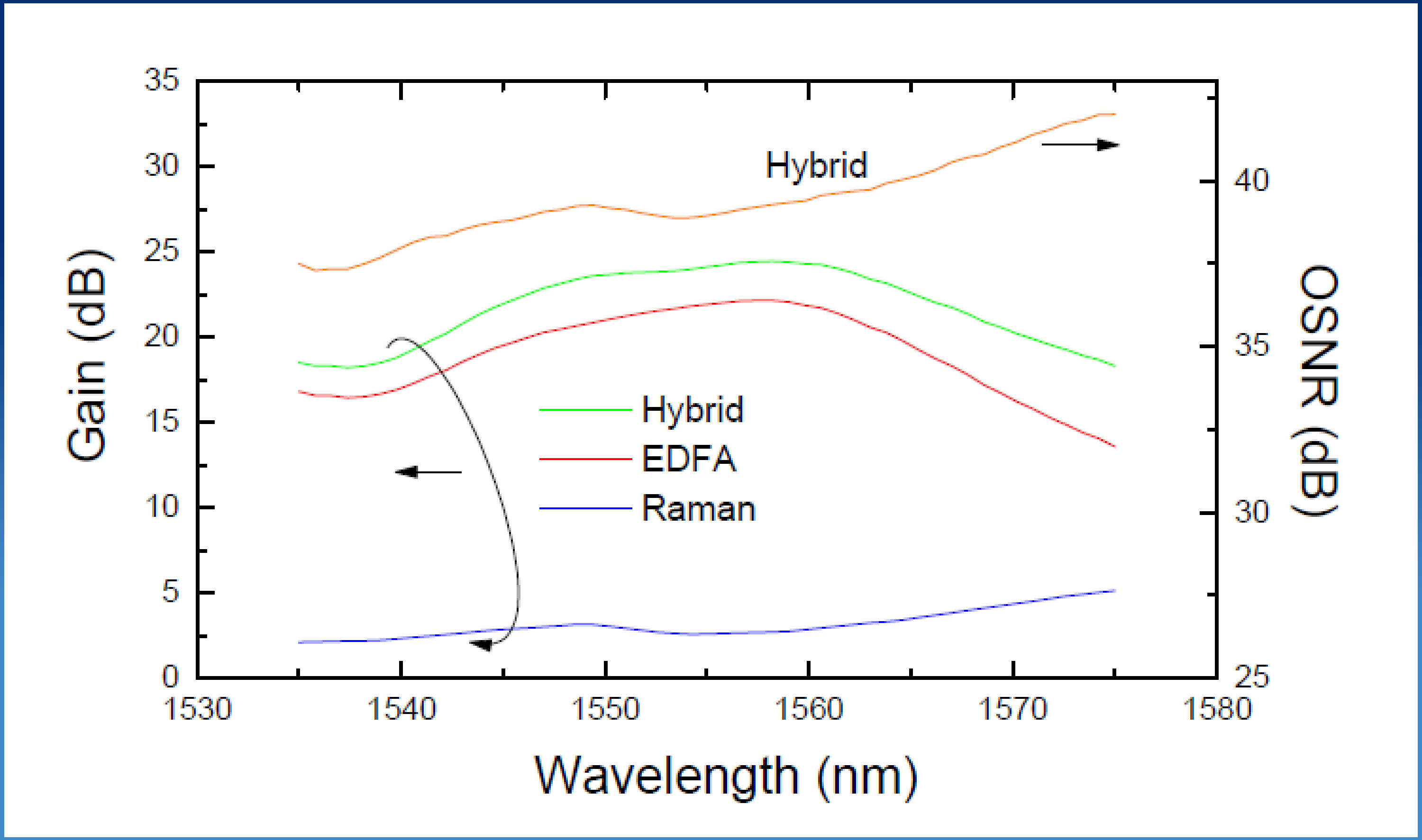
Hybrid Amplifier Layout - copropagating



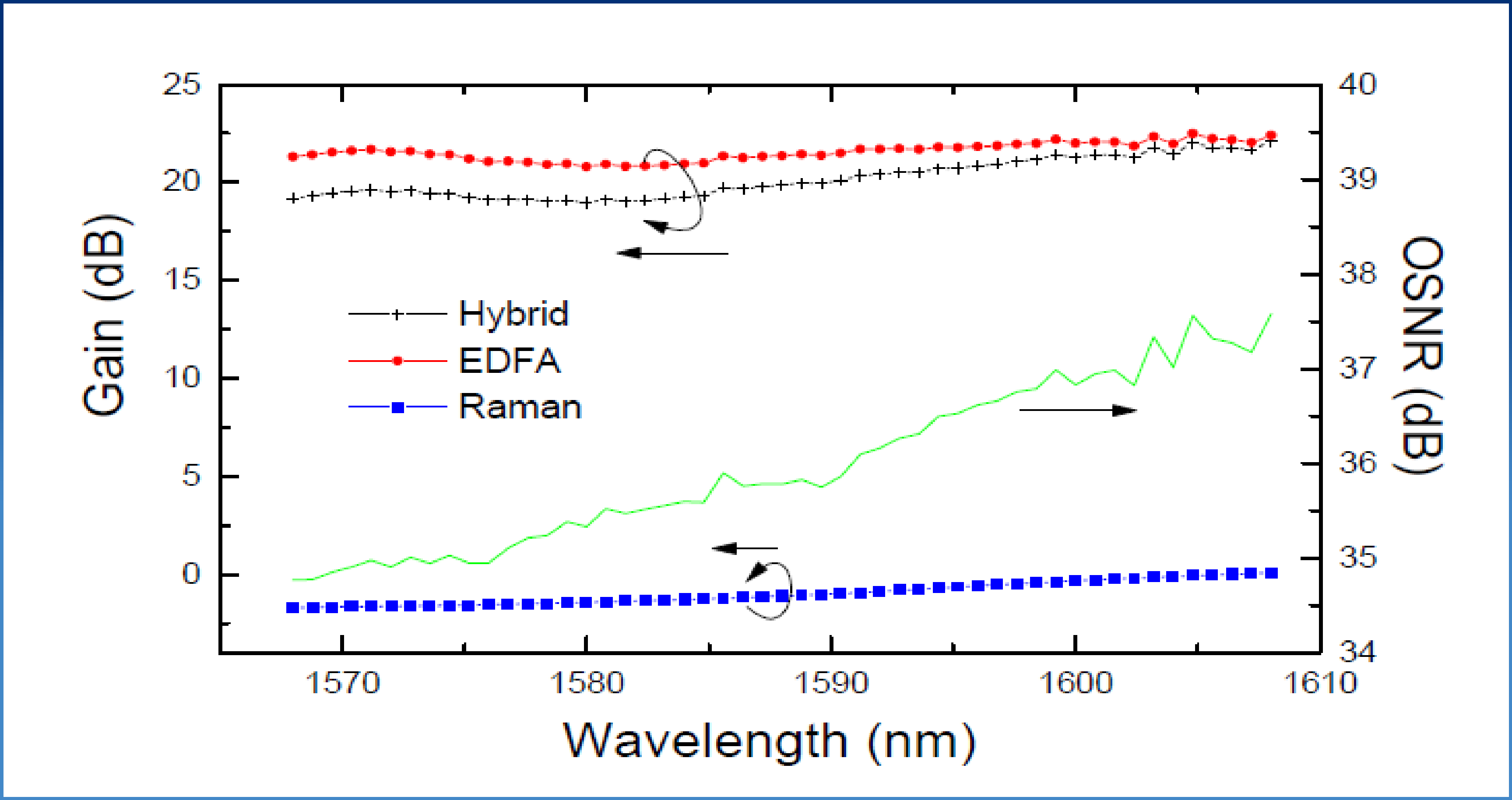
Pumping Power Setting



Evaluating Results in the C-band Wavelength Range

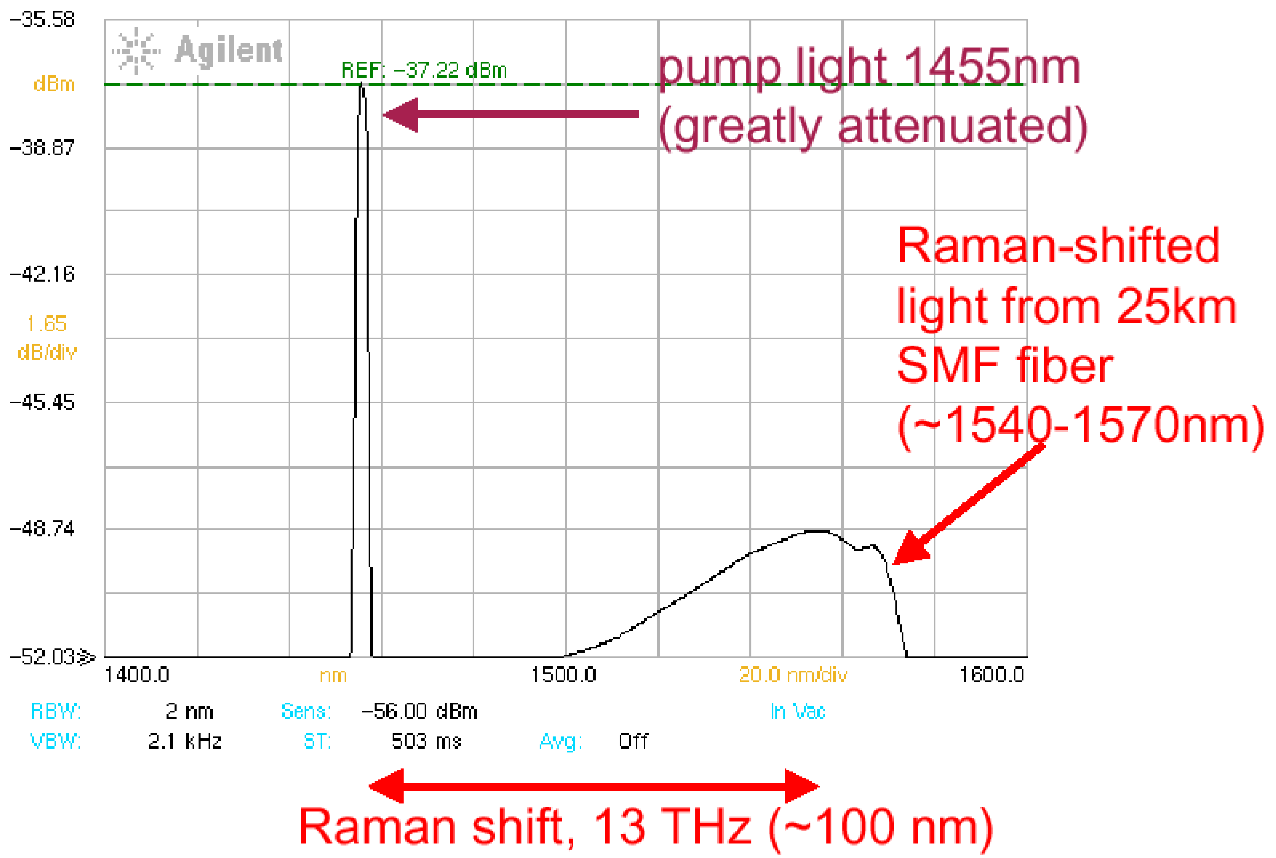


Results in L-band Wavelength Range



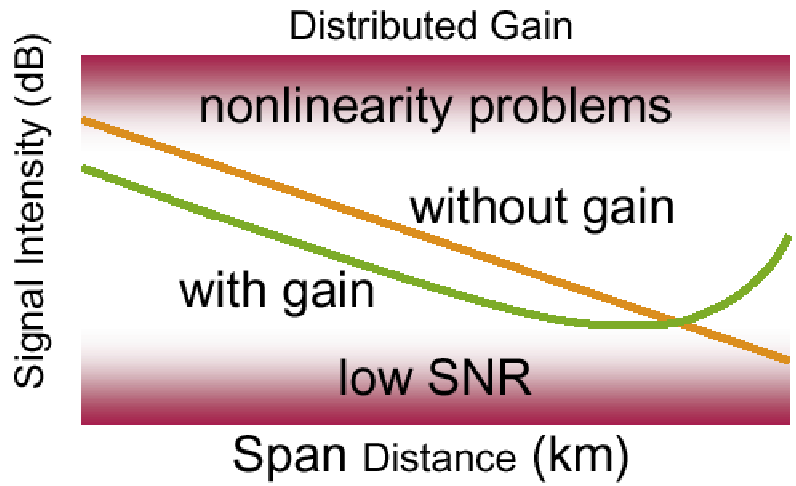
Raman Amplification

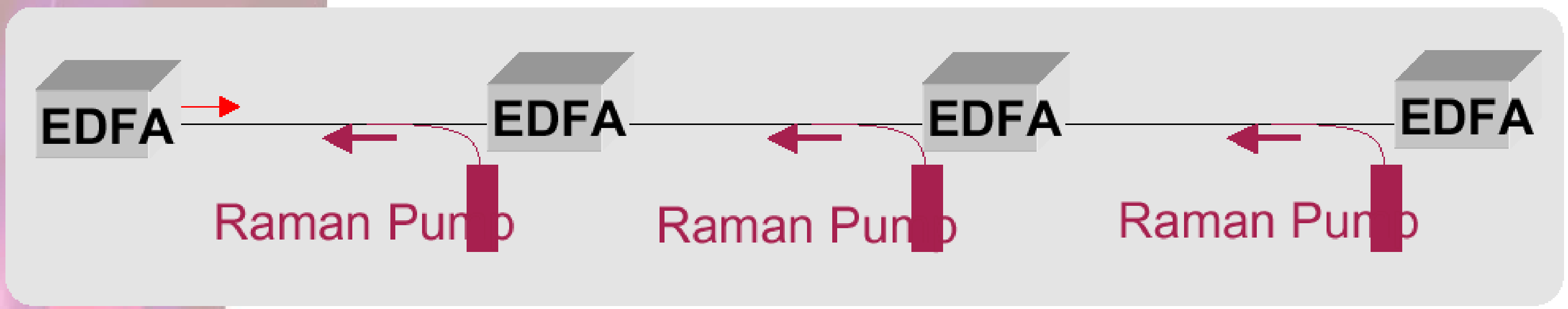
The Raman effect "shifts" some light to longer wavelength



Benefit of distributed Raman amplification

Longer spans possible with lower intensity excursions: Better SNR less fiber non-linearity

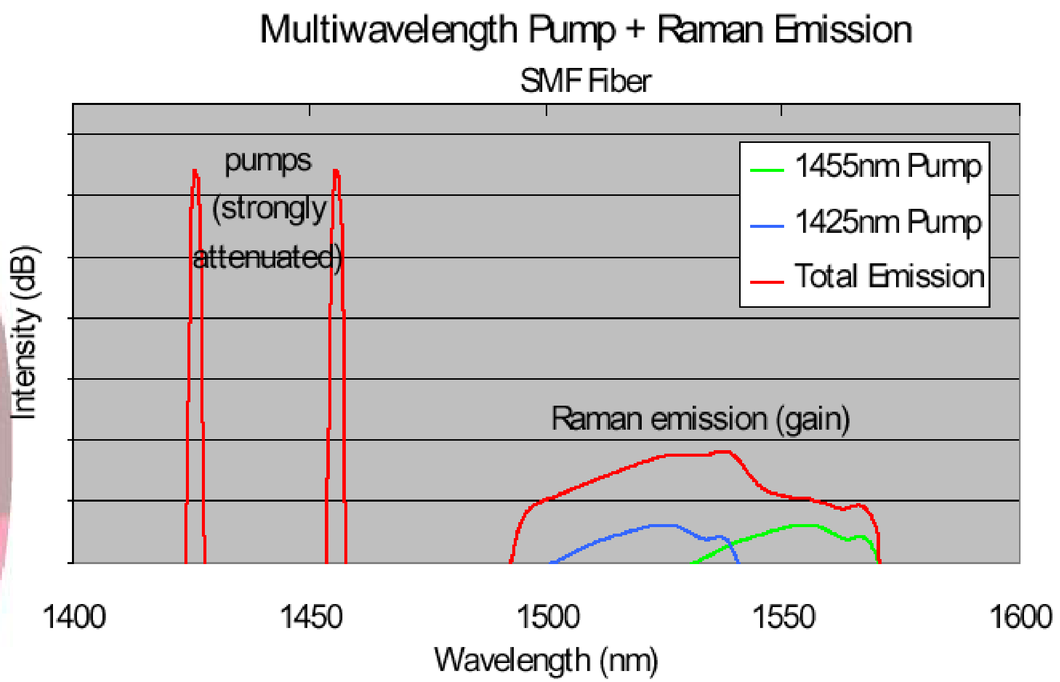




Can be added to EDFA-amplified spans as pre-amplifier! "hybrid amplification"

This helps achieve long distances with a series of amplified spans without expensive optoelectrical regeneration and enables high bit rates like 40

Raman amplification benefit



Benefit:

- Because the gain band moves with the pump wavelength, new bands and broad bands can be amplified!
- Pump intensities can be adjusted for gain flatness!

Future Amplifier:

Extending from C+L to S,E band

Trends in optical amplifiers

- EDFA vs. Raman -

- **EDFA**: Mature technology
  - **New materials (Fluoride, Tellurite)**
  - **New dopant (Pr, Tm) ~PDFA, TDFA**
    - ***to exhibit broader and flatter gain***
- **Raman amplifier**: Advantage in long-haul (LH) space
  - SN improvement by distributed Raman
  - Flat gain by multiple pump wavelength

>> ***<u>Efficiency merit of EDFA is offset by required gain flattening.</u>***

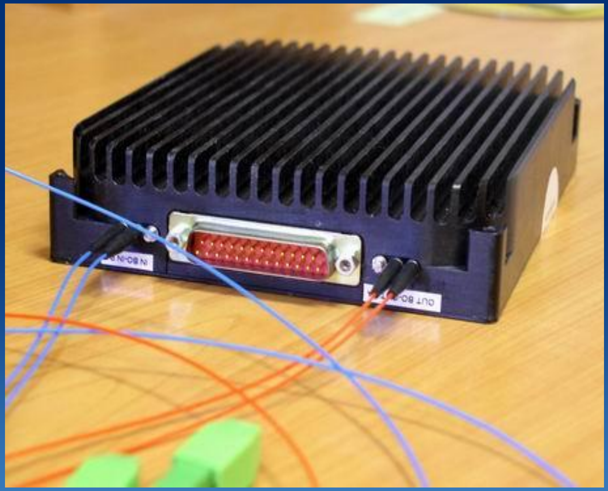
>> ***<u>Raman systems are challenging EDFA stronghold in LH applications.</u>***

Optical amplifier type

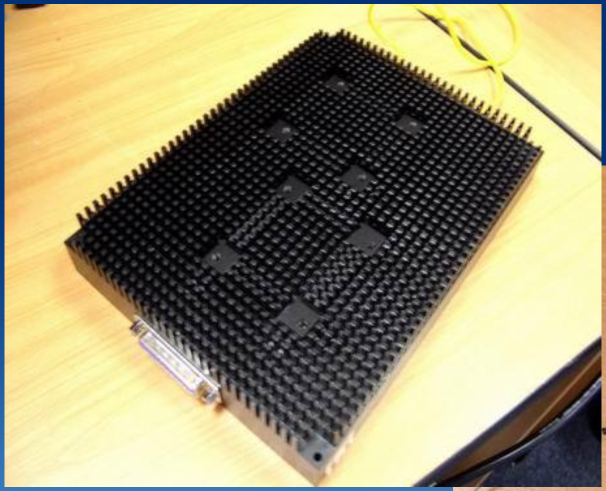
- **Rare earth-Doped Fiber Amplifiers**
  - Erbium-Doped Fiber Amplifiers (EDFA) : C, L-Band
  - Thulium-Doped Fiber Amplifiers (TDFA) : S-Band
  - Praseodymium-Doped Fiber Amplifiers (PDFA) : O-Band
- **Fiber Raman Amplifiers**
  - Discrete Raman Amplifiers
  - Distributed Raman Amplifiers (DRA)
- **Semiconductor Optical Amplifiers** (SOA)
  - conventional SOA
  - GC-SOA (Gain-Clamped SOA)
  - LOA (Linear Optical Amplifier)

Amplifier Examples

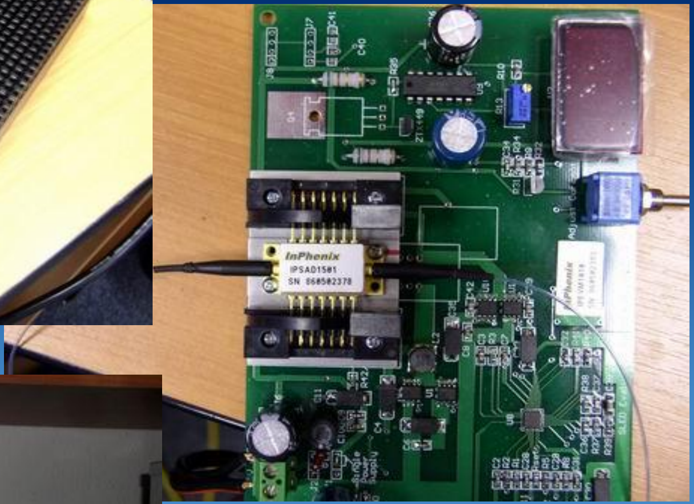
- **EDFA**



- **Raman**

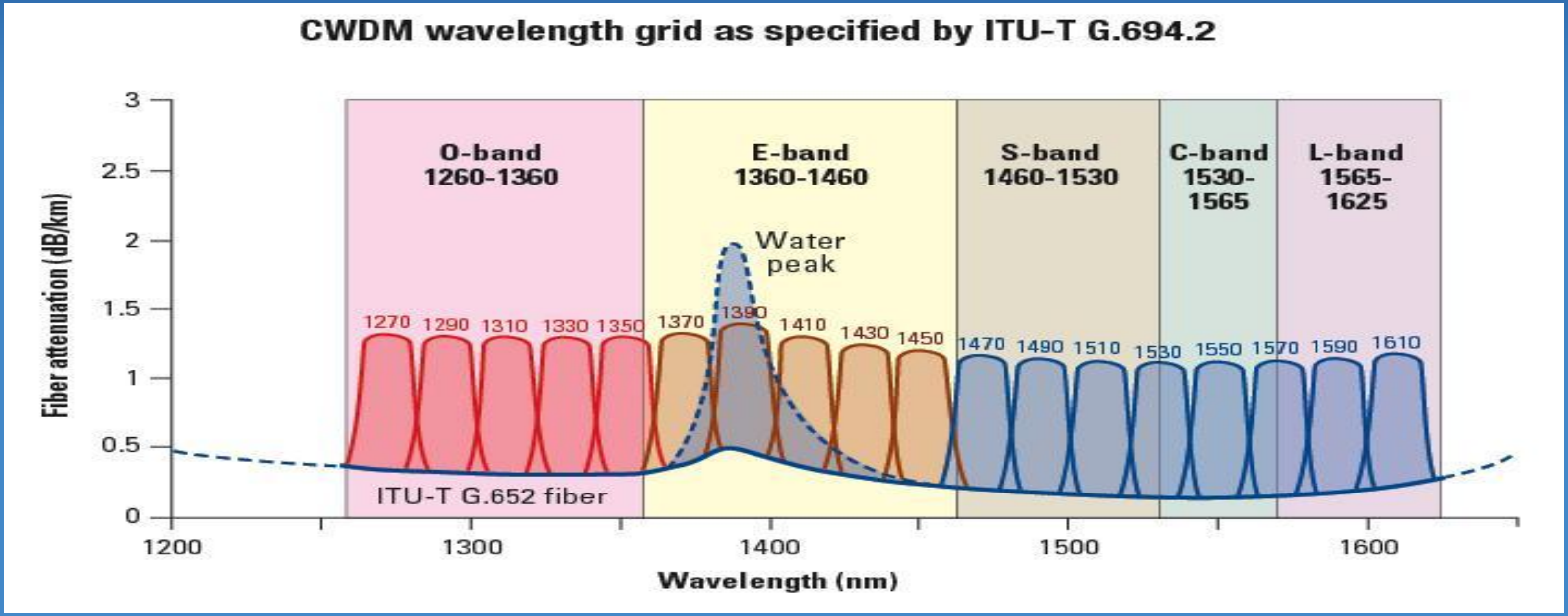


- **SOA**

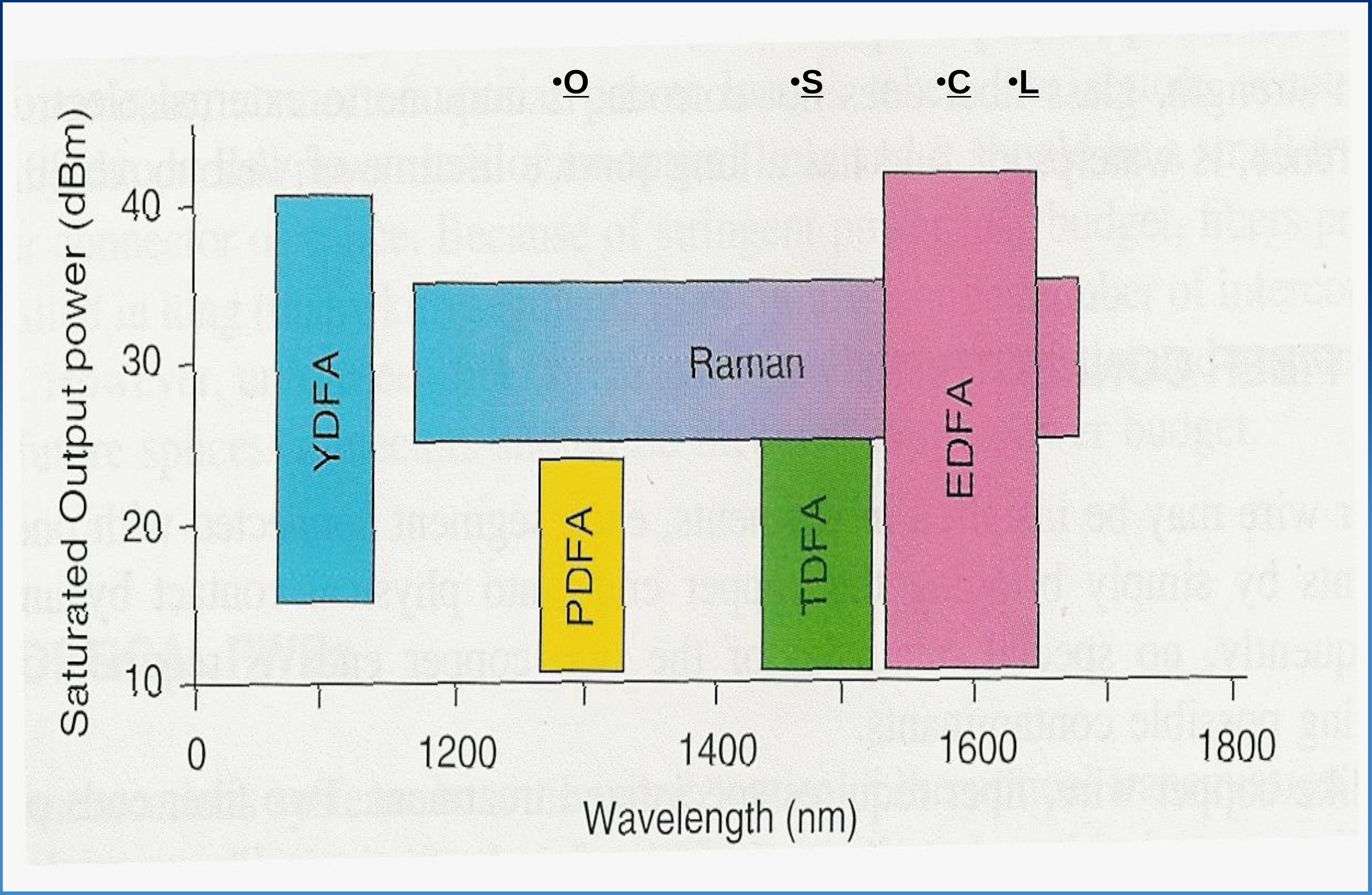


Optical Fibre Amplifiers

- Active environment – special fibre doped with one or more rare earth element (Er, Nd, Pr, Tm,…or combination Er/Yb, Tm/Yb,..)
- PDFA
  - Pr doped, suitable for (1280-1340nm)
  - G>30dB, Pout >16 dBm, NF <7dB
- TDFA
  - Tm doped, suitable for (1440-1520nm)
  - G>30dB, Pout >20 dBm, NF <7dB



Spectral usability of Amplifiers

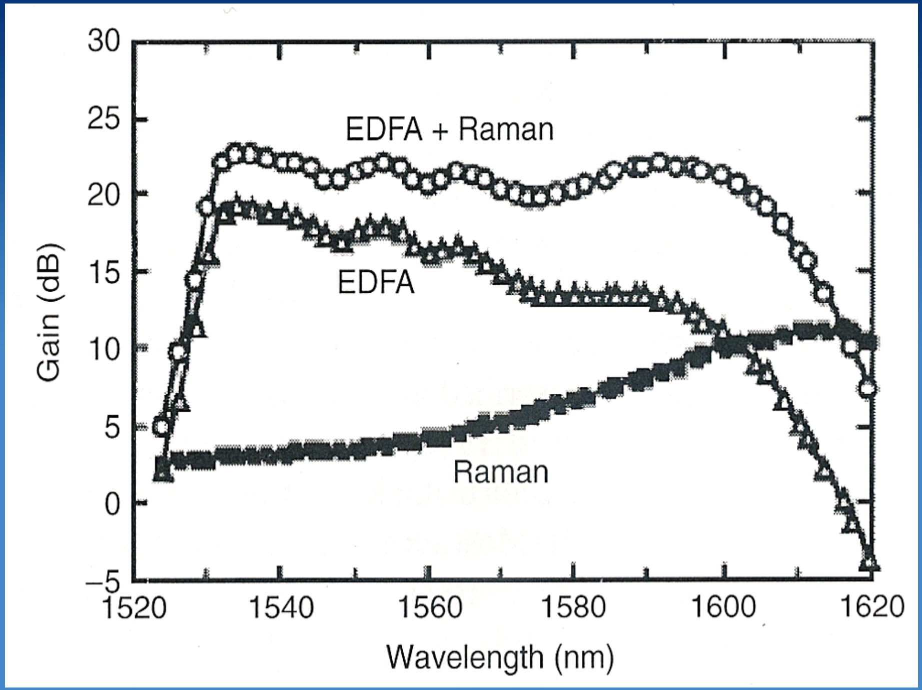


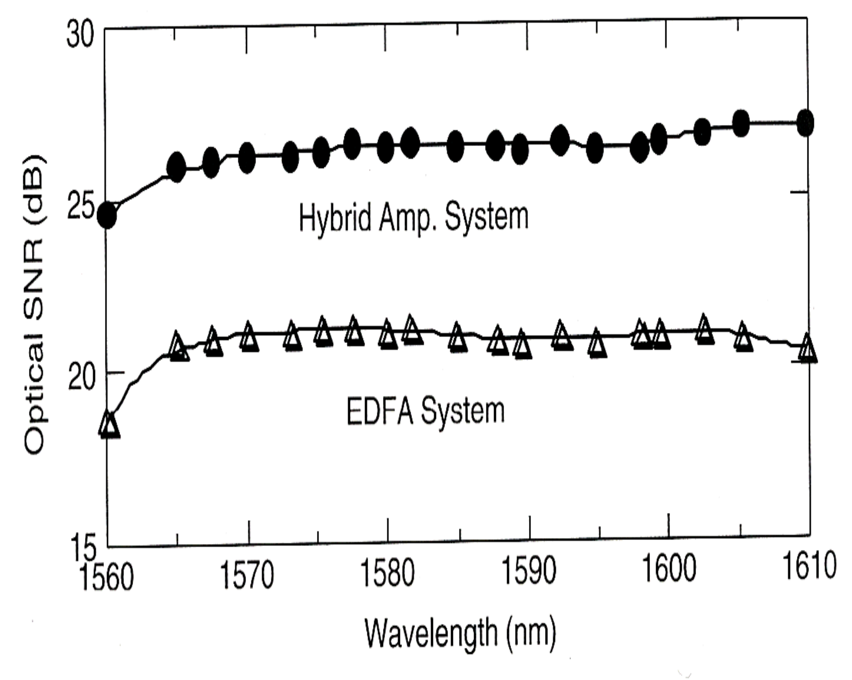
Amplifiers for (D)WDM I

- EDFA – needs gain flattening
  - Glass composition (F, Te glass host)
    - Single stage EDFA, silica host: bandwidth 15nm
    - Single stage EDFA, fluoride host: bandwidth 25nm
  - Equalizers
    - Two stage, silica or fluoride host, no gain flattening: 30nm
    - Two stage, silica host, gain flattening: 50nm
    - Two stage, tellurite host, gain flattening: 80nm
  - Hybrid OA
    - Multi-arm - two band operation, silica host: 85nm

Amplifiers for (D)WDM II

- Hybrid EDFA/Raman
  - Bandwidth can be tailored ~80nm
  - Lower NF than EDFA separate!!





OSNR improvement

Rare earth (Er, Tm, Pr) -Doped Fiber Amplifiers

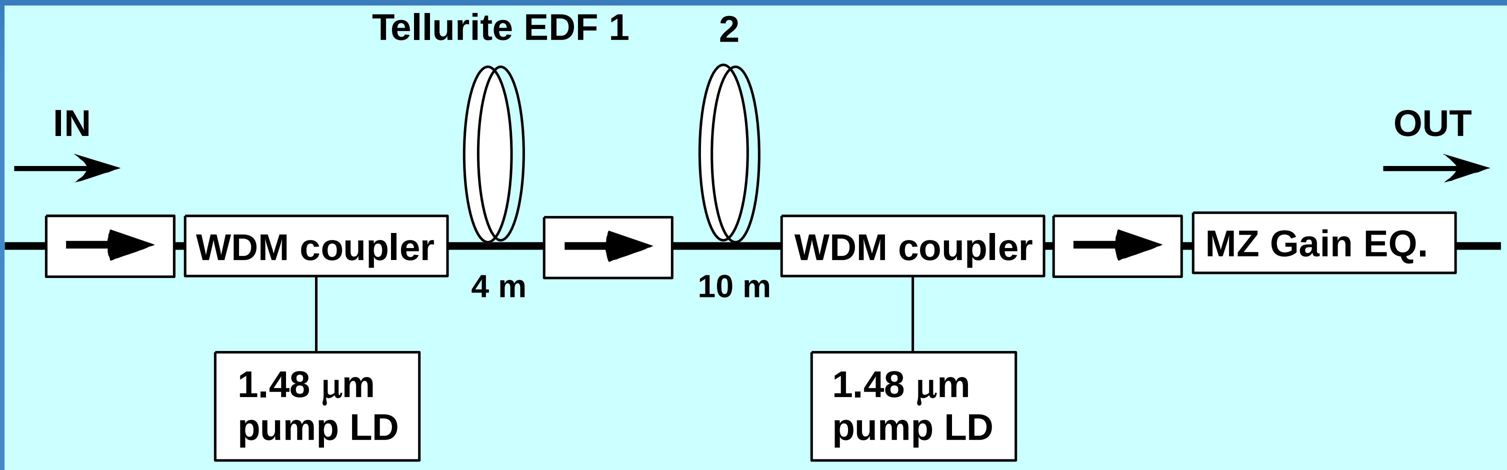
- **Gain band**:

  Er (C, L-Band), **Tm** (S-Band), **Pr** (O-Band)

  76 nm (1532-1608 nm) record gain bandwidth in single band configuration [M. Yamada et al.,OFC'98PD].

  - Flat gain: 21 dB, Noise figure: 7 dB

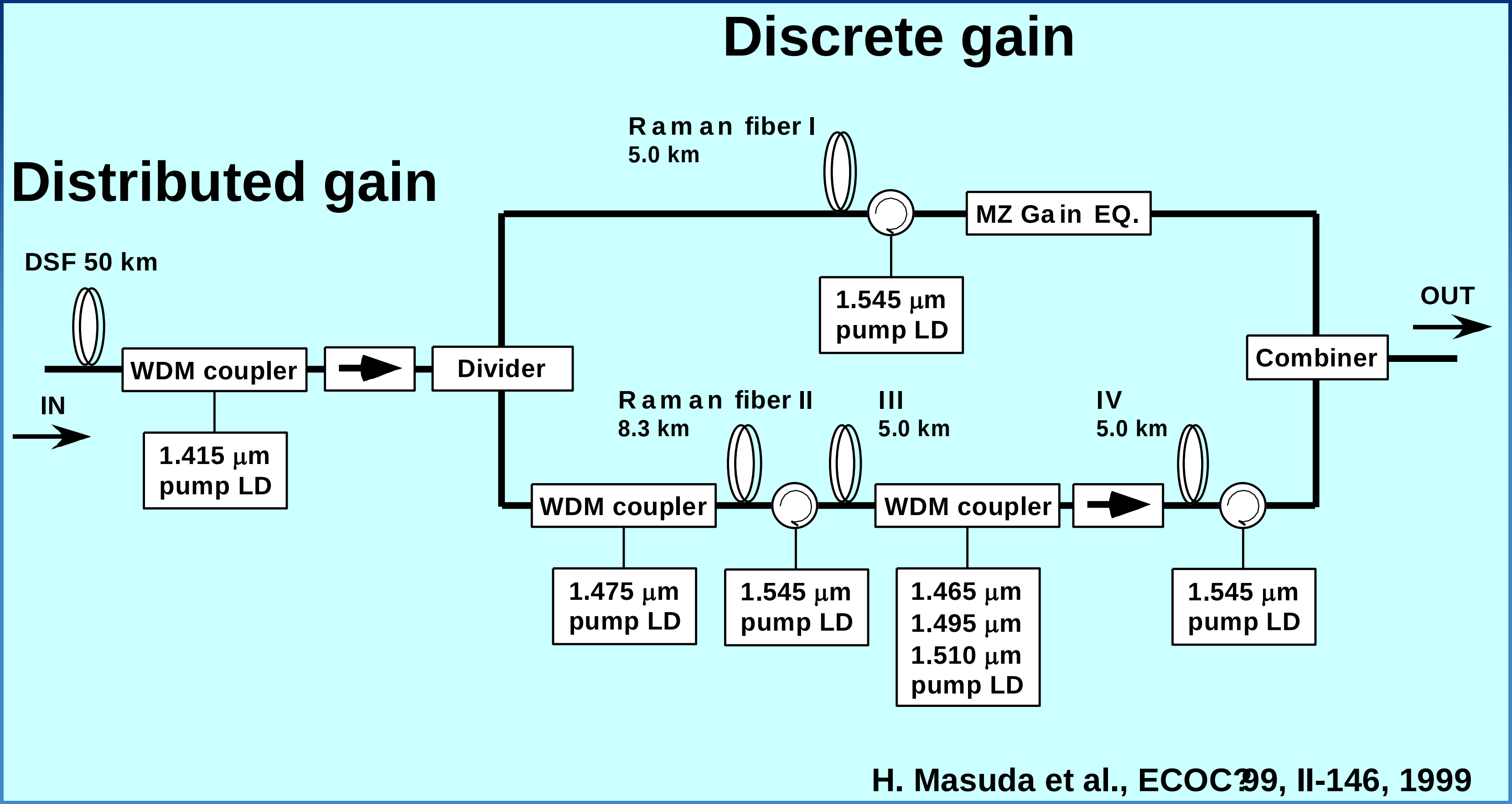
  - Gain equalizer: two MZ filters with FSR of 32 and 120 nm



M. Yamada et al., OFC'98 PD, PD7, 1998

Fiber Raman Amplifiers

- **Gain band: 1.3~1.7 um** (tunable by pump wavelength)
- **132 nm record gain bandwidth in double band configuration** has been achieved [H. Masuda et al., ECOC'99].
  - Combination of **Distributed Raman amplifiers (DRA) and discrete Raman**
  - Two-gain-band Raman amplifier



Hiroji Masuda's work (NTT)： Distributed v.s. Discrete

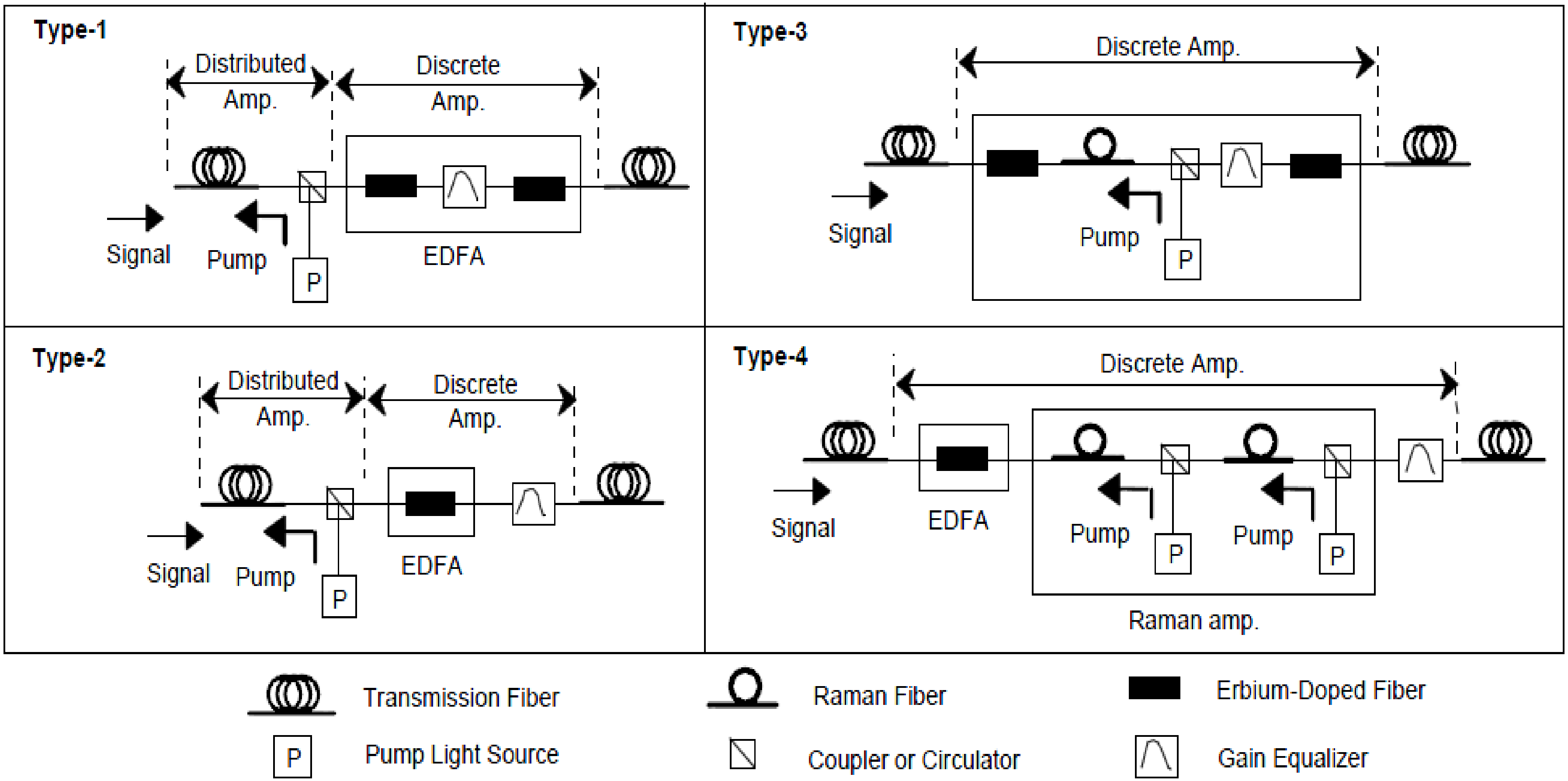


Fig. 1 Four amplifier configurations

Hiroji Masuda's work (NTT)

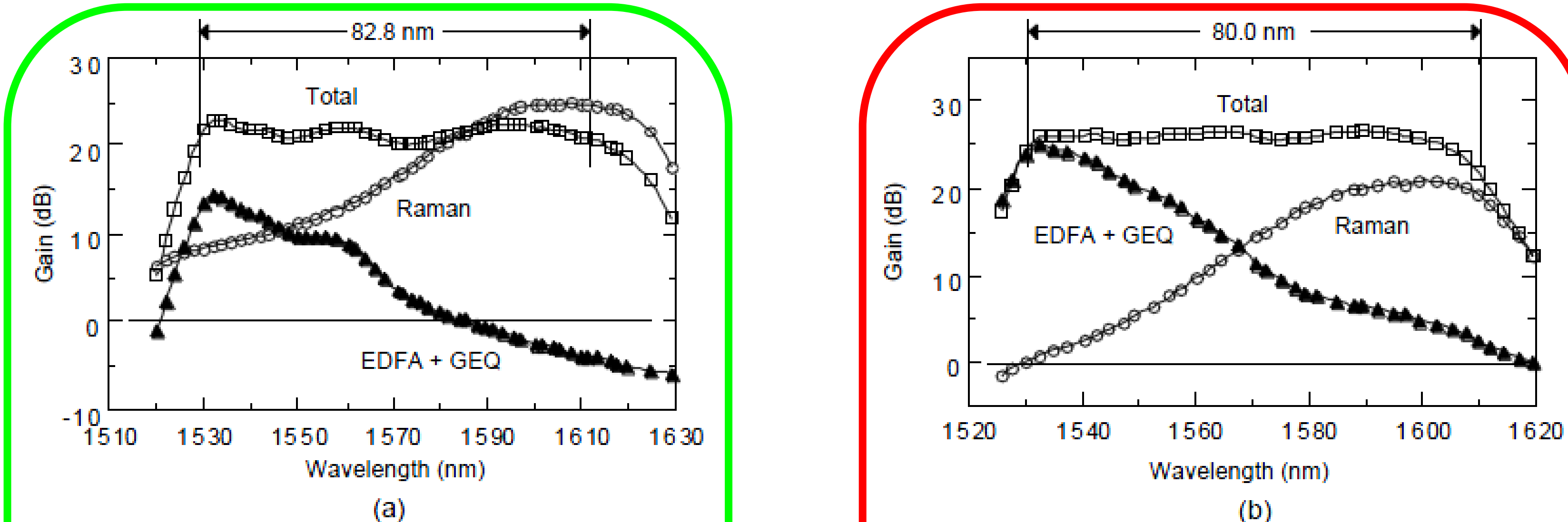
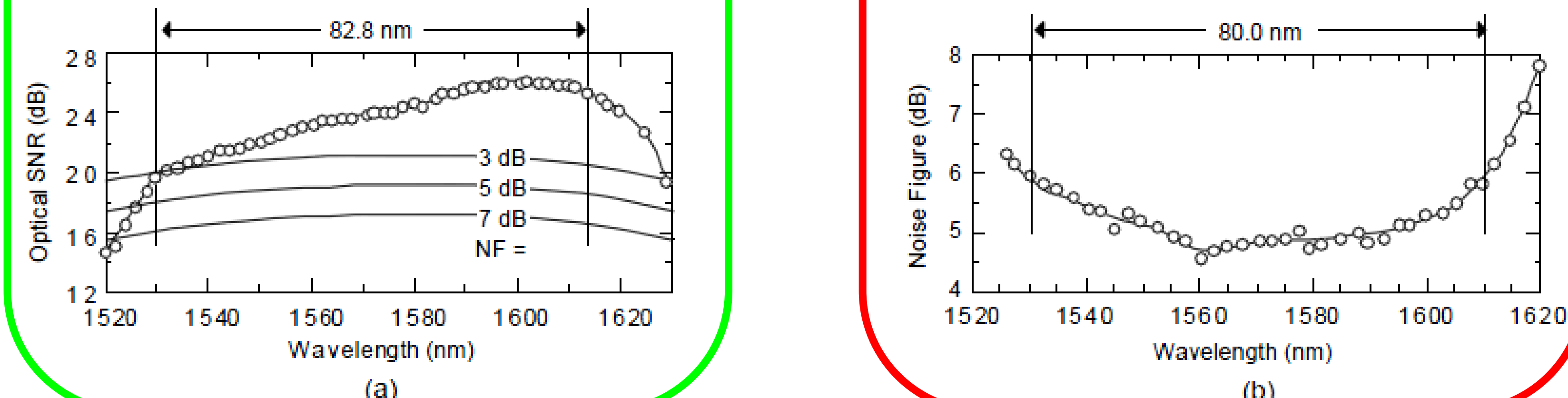


Fig. 2 Gain spectra of wideband hybrid amplifiers
(a) Type-2 amplifier, (b) Type-4 amplifier



Type 2

Fig. 3 (a) Optical SNR spectrum for Type-2 amplifier (Fig. 2
(b) Noise figure spectrum for Type-4 amplifier (Fig. 2(b))

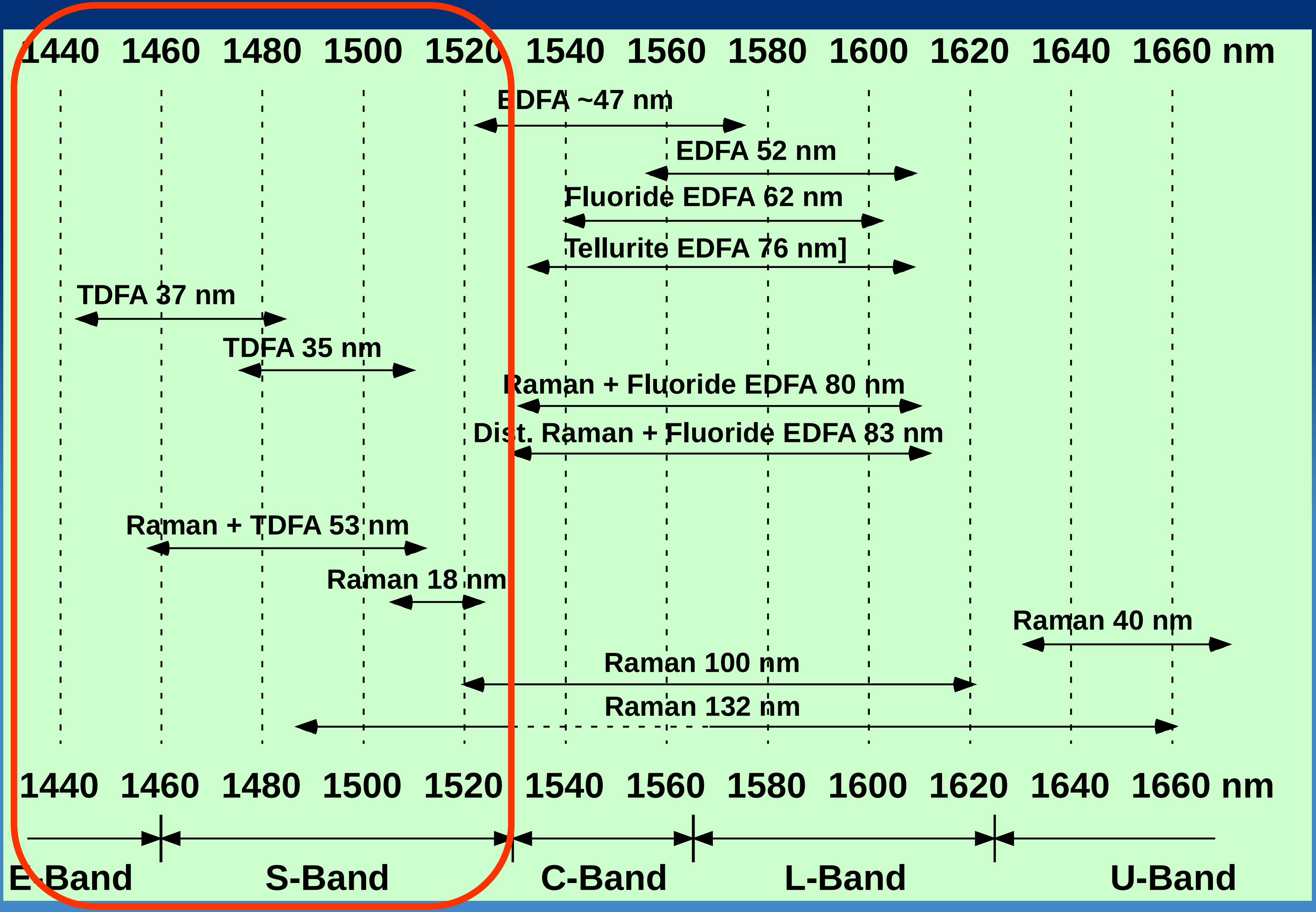
Type 4

Hiroji Masuda's work (NTT, 2000)

- BER performance of the Type-2
  - tested in a 9 x 2.5 Gb/s WDM
  - The signal power launched 5 dBm/total = - 4.5 dBm/channel.
  - Error free operation with BERs under 10^{-11}
  - 3-dB gain-bandwidths of up to 82.8 nm
  - gain > 20 dB
  - noise figure of 3 dB

- Type-4 amplifier:
  - measured in a 15 x 2.5 Gb/s WDM transmission experiment
  - The output signal power from the Type-4 discrete amplifier was as high as 13.8 dBm/total = 2 dBm/channel.
  - The discrete type amplifier (Type-4) yielded low noise figures under 6 dB, and the distributed Raman type amplifier (Type-2) yielded optical SNRs higher than those of the discrete amplifier system with an amplifier.

Hiroji Masuda's work (NTT, 2002)



Hiroji Masuda's work (NTT, 2002) Ultra broadband

- high concentration TDFs to shift the amplifier's spectral gain
- double-pass reflective configuration with a circulator and a mirror and a one color 1.4-μm pumping scheme in the second stage
- to obtain a high ε of 42%
- The gain-bandwidth Δλ with gains of more than 26-dB was 30 nm (1480-1510 nm)
- NFs were less than 7 dB in the gain-band.

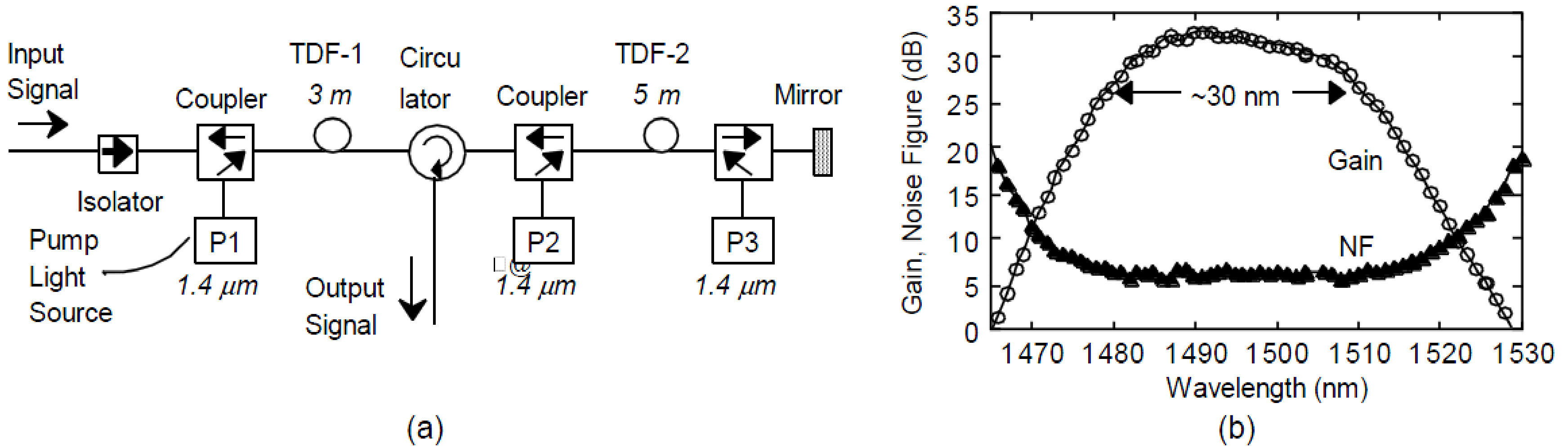
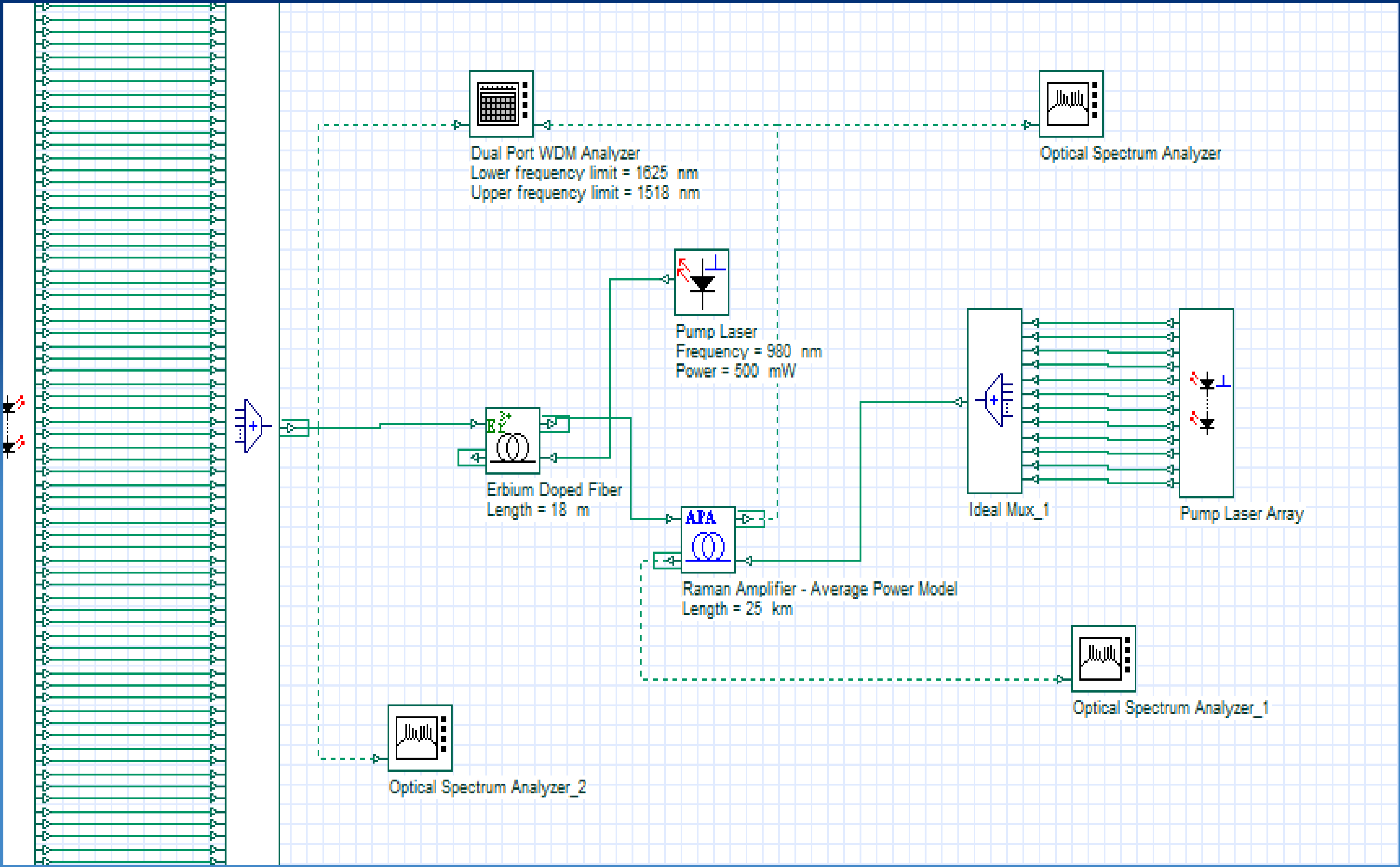
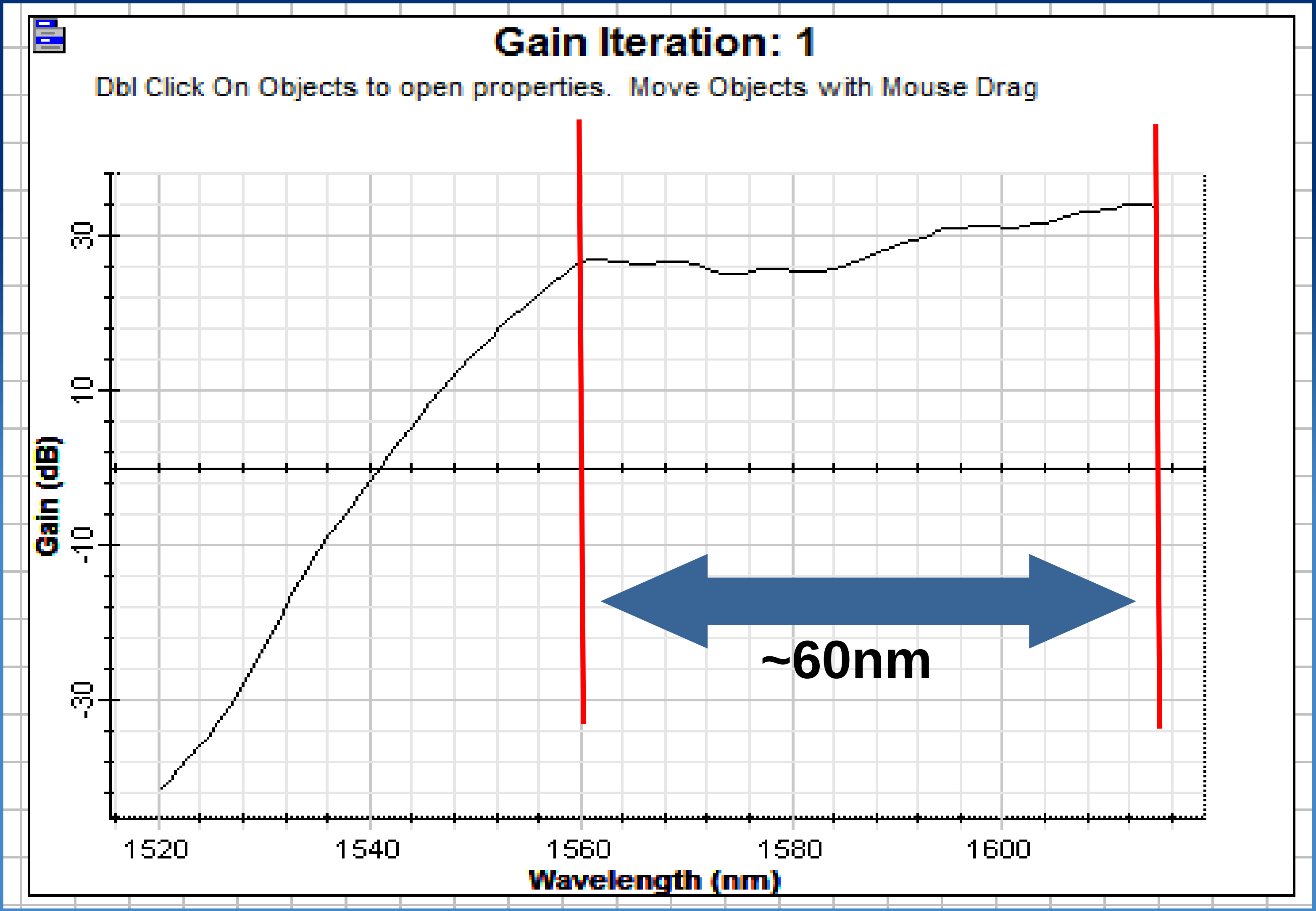


Fig. 2 Highly efficient gain-shifted TDFA. (a) Amplifier configuration, (b) gain and noise figure spectra

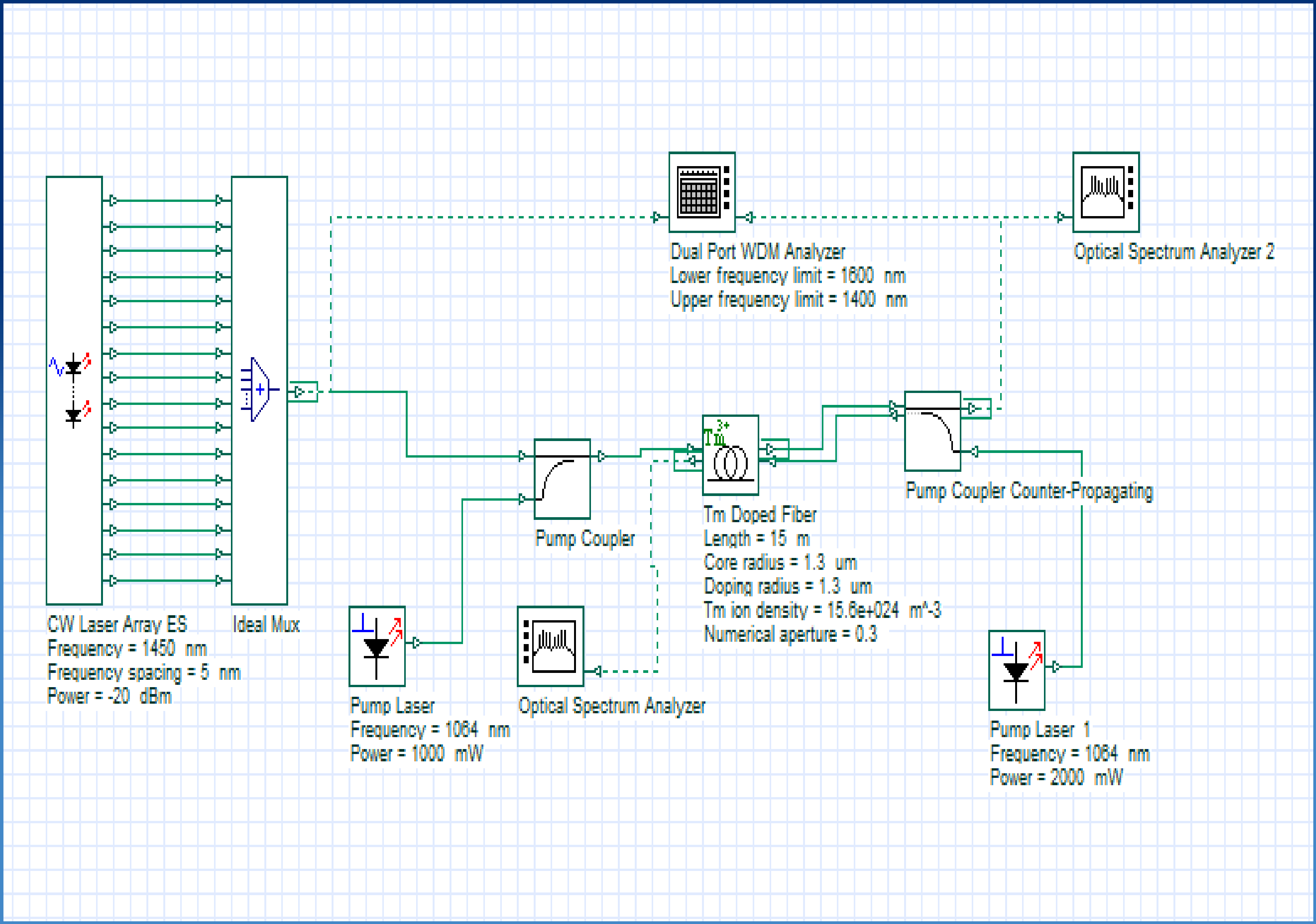
EDFA+8 wavelength Raman



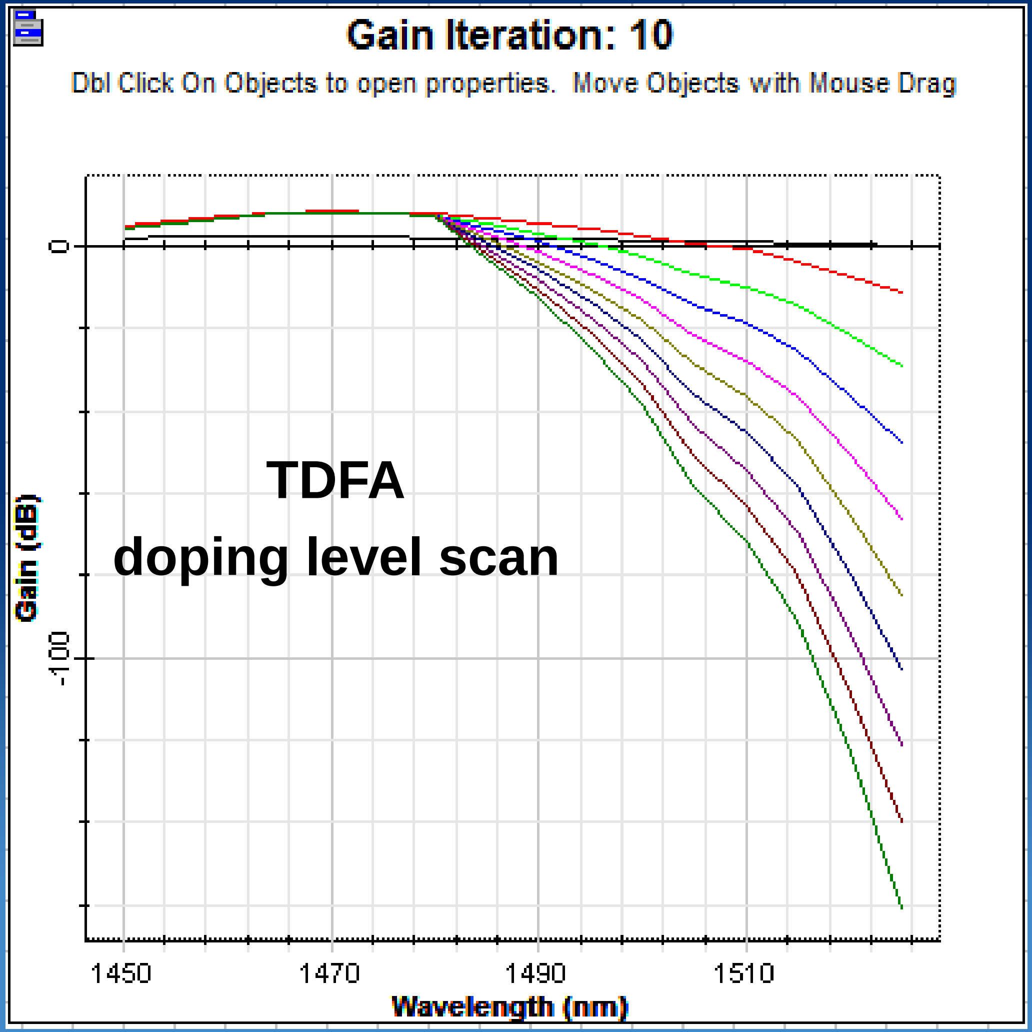
After a few trial and error



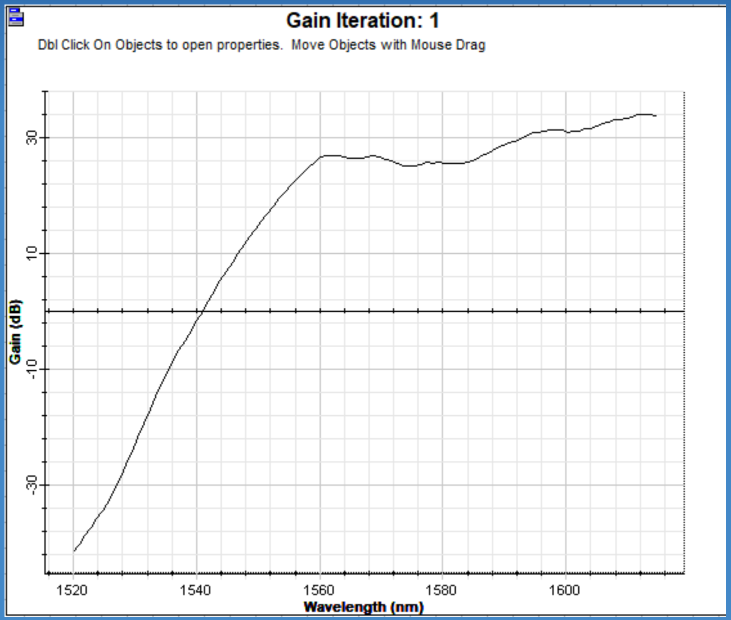
TDFA



Dual Port WDM Analyzer
Lower frequency limit = 1600 nm
Upper frequency limit = 1400 nm
Optical Spectrum Analyzer 2
Pump Coupler Counter-Propagating
Pump Coupler
Tm Doped Fiber
Length = 15 m
Core radius = 1.3 um
Doping radius = 1.3 um
Tm ion density = 15.8e+024 m^-3
Numerical aperture = 0.3
CW Laser Array ES
Frequency = 1450 nm
Frequency spacing = 5 nm
Power = -20 dBm
Ideal Mux
Pump Laser
Frequency = 1064 nm
Power = 1000 mW
Optical Spectrum Analyzer
Pump Laser 1
Frequency = 1064 nm
Power = 2000 mW



180nm Gain Band!?



In-class Exercise

- Design a hybrid EDFA+Raman to observe >60 nm flattened gain band, using 4-8 pumping wavelengths, and keeping in your mind how big the gap is between pumping and gain
- Design your own TDFA with both counter-propagation and co-propagation pumping
- Parameters:
  - Pumping wavelength:1064nm
  - Pumping power: >1W
  - Fiber length: ~10-20m